# Md. Ridwanul Islam Antu

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## EDUCATION

**Military Institute of Science and Technology (MIST)** *Dhaka, Bangladesh*
M.Sc. in Mechanical Engineering *Apr 2024-Ongoing*

**Military Institute of Science and Technology (MIST)** *Dhaka, Bangladesh*
B.Sc. in Mechanical Engineering *Jan 2020-Mar 2024*

## RESEARCH INTERESTS

- Robotics
- Autonomous Systems
- Digital Twin
- Design Optimization
- Machine Learning
- Additive Manufacturing
- Mechatronics
- Drone Design

## PUBLICATIONS

**Antu, M. R. I., Rahman, M. H.** "Design, Development, and Workspace Evaluation of a Ceiling-Mounted Four-Axis SCARA Robotic Arm for Light-Duty Automation." *Accepted at the IEEE International Conference on Robotics, Automation, Artificial-Intelligence and Internet-of-Things (RAAICON 2026)*

- Derived a DH-based kinematic model with closed-form forward/inverse kinematics for the P–P–R–R chain.
- Quantified a 110.4% planar workspace gain over an equivalent fixed-base SCARA via a custom Python reachability analysis.

**M. R. I. Antu, M. H. Rahman**, "Kinematic Performance Analysis and Dimensional Synthesis of a 4-DOF Ceiling-Mounted Rail-Augmented SCARA Manipulator," **(manuscript in preparation.)**

- Built a ROS 2 digital twin from the prototype CAD that matches the analytical kinematics and Jacobian to machine precision over 2,640 test configurations.
- Derived the Jacobian and proved that all singularities lie on the workspace boundary; optimized the link ratio for 63 % more oriented-grasp area (0.638 to 1.040 m²).

## RESEARCH EXPERIENCE

**1. Hyper-Redundant Oblique Swivel Joint Manipulator**
*Supervisor: Dr. Md Sayem Hossain Bhuiyan* *Oct 2026 – **Ongoing***

- Designed an 8-DOF hyper-redundant arm (base yaw plus seven oblique swivel joints, β = 22.5°) in SolidWorks.
- Building a ROS 2 simulation framework (URDF/xacro, RViz) with Jacobian null-space IK for confined-space reach.

**2. ML-Enabled Predictive Digital Twin Framework for Smart Grid Demand Forecasting in Bangladesh**
*Postgraduate Thesis, Supervisor: Dr. Md Sayem Hossain Bhuiyan* *Apr 2025 – **Ongoing***

- Built a forecasting pipeline on 10 years of hourly PGCB national demand data merged with multi-city meteorological records.
- Diagnosed a distribution-shift failure in tree-based models caused by sustained demand growth; resolved via a trend-normalized target reformulation.
- Applied SHAP to quantify demand-history vs. meteorological feature contributions to forecasts.
- Designing an offline scenario-simulation framework for grid decision support, scoped distinctly from real-time digital-twin control architectures.

**3. Study on Minimizing Sound Signature of Quadcopter Propeller**
*Undergraduate Thesis, Supervisor: Dr. Muammer Din Arif* *Apr 2023 – Mar 2024*

- Studied multiple propeller types, focusing on tri-toroidal and modified conventional designs to reduce noise while maintaining comparable thrust performance.
- Conducted CFD simulations to analyze airflow and noise in ANSYS (Fluent) using the Multiple Reference Frame (MRF) approach and validated results experimentally.
- Modified conventional tri-blade showed the least difference (17.25%) between simulation and experiment.

**4. Design, Development, and Fabrication of a Ceiling Mounted Multipurpose SCARA Robotic Arm**
*Integrated Design Project, Supervisor: Shah Md. Ahasan Siddique* *Apr 2022 – Mar 2023*

- Developed an Arduino-based ceiling-mounted four-axis SCARA robotic arm for precise multi-axis movement.
- Designed in SolidWorks, integrating 3D-printed joints and aluminum/SS components for enhanced stability.
- Achieved improved stability and workspace efficiency by removing the ground base.

## PROJECTS

**1. Design and Manufacture of Mars Rover, PHOENIX 3.0**

*MIST Mongol Barota, Anatolian Rover Challenge 2023* *Nov 2022 – Jul 2023*

***Chassis & Wheel System***

- Developed a steerable wheel subsystem with stepper motor and thrust bearings for precise maneuverability.
- 3D-printed honeycomb structured non-pneumatic tires using TPU for better shock absorption.

***Suspension System***

- Designed a rocker bogie mechanism to maintain stability and absorb frontal and rear impacts on uneven terrain.
- Implemented a bevel gear differential system for efficient torque distribution and compact design.
- Fabricated carbon fiber rocker arms to increase strength, reduce weight, and improve suspension durability.

***5 Degrees of Freedom (DOF) articulated arm***

- Inverse kinematics for precise movement with 5:1 spur and worm gears for waist motion and tight grip.
- Carbon fiber links reduce weight and actuator enhances pulling power.
- Bevel gear enables dual motion with custom bearings and 3D-printed spacers for joint stability.

**2. Machine Learning-Based Path Optimization for Mobile Robots (Simulation Study)** *2025*

- Built a Python-based ML model to predict optimal robot paths using A* simulation data.
- Trained Linear Regression and Random Forest models to evaluate path cost prediction accuracy.

## LEADERSHIP EXPERIENCE

**1. MIST Mongol Barota**

*Mechanical Team Lead* *Apr 2023 – Mar 2024*

- Led the mechanical team at ARC'23 in Ankara, Turkey, overseeing rover design, fabrication, and integration.
- Collaborated with interdisciplinary teams to ensure successful mechanical - electrical integration.

**2. MOTOMIST Automotive Club**

*Club President* *Apr 2023 – Mar 2024*

- Supervised all technical and organizational operations, mentoring teams in competitive project development.
- Organized technical workshops, competitions and seminars.

## WORK EXPERIENCE

**1. NKDM Association (NK-NK India-DMRC-MOTT UK-MOTT India-DDC)**

*Junior Engineer | Dhaka MRT Line 6 Project* *Dec 2024 – Present*

- Contributed to technical review, documentation, and site supervision for Rolling Stock and Depot Equipment.
- Coordinated with multidisciplinary teams for data-driven progress analysis and compliance assessment.

**2. Walton Hi - Tech Industries PLC**

*Industrial Trainee | Engineering Services Management* *Jan 2023 - Feb 2023*

- Observed large-scale manufacturing operations and quality control processes for refrigerators.
- Gained exposure to metal casting, injection molding, and utility system management.

## SKILLS

1. **Programming**: Python, C, MATLAB
2. **Robotics & Control:** Arduino**,** ROS2, Gazebo, RViz2, MoveIt2
3. **CAD and Simulation**: SolidWorks, AutoCAD, Ansys (fluent), COMSOL
4. **Manufacturing:** 3D Printing, Ultimaker Cura
5. **Documentation:** MS Office, Aconex, LaTeX
6. **Language:** Bangla, English **(IELTS: Overall Band- 7, L-8, R-6.5, W-6.5, S- 7)**

## ACHIEVEMENTS AND HONORS

1. Anatolian Rover Challenge 2023 – 5th position (Team Achievement) *2023 (Turkey)*
2. International Rover Design Challenge 2023-16th position (Team Achievement) *2023(Online)*
3. University Rover Challenge 2023 – Finalist (Team Achievement) *2023(Online)*
4. University Rover Challenge 2021 – Champion (Team Achievement) *2021(Online)*
5. Casessination 2.0, Global business case competition – 4th position (Team Achievement) *2021 (Bangladesh)*

## CERTIFICATIONS

1. Supervised Machine Learning: Regression and Classification - **Stanford University (Coursera)** *2025*
2. Certified SOLIDWORKS CAD Design Associate (CSWA), **Dassault Systemes** *2025*

**3.** Anatolian Rover Challenge 2023, **uket Space Exploration Society, Turkey** *2023*